MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

CLOUD COMPUTING COURSE

Build Skills. Gain Hands-on Experience. Grow Your Career.



COURSE HIGHLIGHTS

- Linux Administration
- AWS Cloud Computing
- DevOps & CI/CD
- Git, Jenkins & Ansible
- Docker & Containerization

CLOUD COMPUTING MASTER PROGRAM

Learn • Build • Deploy • Grow

COURSE OVERVIEW

About the Course

Build a strong foundation in Linux, Cloud Computing and DevOps through a structured learning journey from system administration to cloud deployment, automation and containerization.

This program covers Linux Administration, AWS Cloud Services, Git, DevOps, Jenkins, Ansible, Maven, Docker and CI/CD automation.



BUILD YOUR SKILLS FOR A STRONGER CLOUD CAREER

WHAT WILL YOU LEARN?

LINUX ADMINISTRATION

- Linux Commands & Shell Scripting
- System Administration
- Networking & Permissions
- Services & Performance Monitoring

AWS CLOUD COMPUTING

- Compute & Storage Services
- Networking & Security
- Databases & Monitoring
- Cloud Architecture

DEVOPS FUNDAMENTALS

- DevOps Culture & Practices
- CI/CD & Automation
- Infrastructure as Code
- Deployment Strategies

GIT & VERSION CONTROL

- Source Code Management
- Branches & Repositories
- Collaboration
- Development Workflows

JENKINS CI/CD

- Build Pipelines
- Continuous Integration
- Testing & Deployment
- Automation

ANSIBLE AUTOMATION

- Configuration Management
- Application Deployment
- Multi-Server Environments
- Playbooks & Roles

MAVEN BUILD AUTOMATION

- Java Projects
- Dependencies Management
- Builds & Packaging
- Project Lifecycle

DOCKER & CONTAINERS

- Containerization Basics
- Docker Images & Volumes
- Docker Compose
- Deploy & Manage Containers

WHO CAN JOIN?

- Students
- Graduates
- Beginners
- Working Professionals
- Developers
- System Administrators
- Aspiring Cloud & DevOps Professionals

LEARNING OUTCOME

By the end of the program, learners will be able to manage Linux systems, deploy applications on AWS, automate infrastructure, build CI/CD pipelines and work with modern container technologies.

LINUX & AWS CLOUD

LINUX ADMINISTRATION & AWS CLOUD

LINUX ADMINISTRATION

LINUX FUNDAMENTALS

- Linux OS & File System
- Shell Commands
- Directory Structure
- Files & Directories
- Permissions & Ownership
- chmod, chown & chgrp

USER & GROUP MANAGEMENT

- User Creation & Management
- Group Management
- Sudo & Access Control
- File & Directory Permissions
- Sudoers Configuration

PACKAGE MANAGEMENT

- apt Package Manager
- yum & rpm
- Package Dependencies
- Software Repositories
- Installing Software from Source

SYSTEM MONITORING

- top & htop
- vmstat & iostat
- CPU & Memory Monitoring
- Disk Space Management
- df, du, mount & umount
- Log Management & journalctl

NETWORKING & FIREWALLS

- IP Addressing
- DNS & Routing
- Network Interfaces
- ip & ifconfig
- iptables
- firewalld

SERVICES & PROCESSES

- systemctl & service
- Process Management
- ps & kill
- Background & Foreground Processes
- cron & at
- systemd Timers

AWS CLOUD COMPUTING

AWS FUNDAMENTALS

- Cloud Computing
- IaaS, PaaS & SaaS
- Cloud Benefits
- Scalability & Elasticity
- AWS Global Infrastructure
- Regions & Availability Zones
- Edge Locations

AWS COMPUTE

EC2

- Instance Launch & Configuration
- Instance Management
- Security Groups
- Elastic IP
- Auto Scaling

Load Balancing

- Elastic Load Balancer
- Application Load Balancer
- Traffic Distribution

AWS Lambda

- Serverless Computing
- Lambda Functions
- Event-Driven Architecture

AWS STORAGE

S3

- Buckets & Objects
- Versioning
- Lifecycle Policies
- Permissions & Access

EBS & Other Storage

- EBS Volumes
- Snapshots
- EFS
- Glacier
- Storage Gateway

HANDS-ON PRACTICAL LEARNING

REAL-WORLD USE CASES

BUILD CLOUD & LINUX EXPERTISE

BOOST YOUR CAREER

SKILLS TODAY FOR A SAFER TOMORROW

KEY SKILLS

Linux Administration → AWS Fundamentals → Cloud Services → Security & Monitoring → Career Growth

AWS, DEVOPS & GIT

AWS CLOUD & DEVOPS

3. AWS NETWORKING & SECURITY

AWS VPC
- VPC Architecture
- Public & Private Subnets
- Route Tables
- Internet Gateway
- NAT Gateway
- Security Groups
- Network ACLs

AWS Connectivity
- Elastic IP
- VPN Connections
- AWS Direct Connect
- Network Security

AWS SECURITY

IAM
- Users & Groups
- Roles & Policies
- Permissions
- Multi-Factor Authentication

Security Services
- AWS KMS
- Encryption & Key Management
- AWS WAF
- AWS Shield
- Cloud Security Best Practices

4. AWS DATABASES & MONITORING

AWS DATABASE SERVICES
- Amazon RDS
- Database Backup & Replication
- Amazon DynamoDB
- Amazon Aurora
- Amazon ElastiCache
- Amazon Redshift

MONITORING & LOGGING

CloudWatch
- Metrics & Monitoring
- Alarms
- Application Logs

CloudTrail
- API Activity Tracking
- Auditing
- Security Monitoring

AWS Management
- Trusted Advisor
- Cost Explorer
- Cloud Cost Monitoring

DEVOPS

5. DEVOPS FUNDAMENTALS

DevOps Concepts
- DevOps Lifecycle
- Development & Testing
- Continuous Integration
- Continuous Delivery
- Continuous Deployment
- Infrastructure as Code

DevOps Collaboration
- Agile & DevOps
- Jira
- Confluence
- Slack
- Team Collaboration
- Version Control Practices

AUTOMATED TESTING
- Unit Testing
- Integration Testing
- Functional Testing
- Selenium
- JUnit

CI/CD
- CI Pipeline Design
- Automated Builds
- Automated Testing
- Application Deployment
- Blue-Green Deployment
- Canary Releases
- Feature Flags

GIT — VERSION CONTROL

6. GIT FUNDAMENTALS

Git Basics
- Repositories
- Commits
- Branches
- Merging
- Git Workflows
- GitFlow & GitHub Flow

BASIC GIT COMMANDS
- git clone
- git add
- git commit
- git push
- git pull
- git branch
- git merge

COLLABORATIVE DEVELOPMENT
- Pull Requests
- Code Reviews
- Merge Conflicts
- Git Hooks
- Pre-Commit & Post-Commit

ADVANCED GIT
- Rebasing vs Merging
- Git Tags
- Releases
- Stashing Changes
- Resetting Changes

KEY SKILLS

Understand → Build → Secure → Deploy → Collaborate → Grow

BUILD MODERN CLOUD SKILLS

PRACTICAL HANDS-ON LABS

BUILD SECURE & SCALABLE SYSTEMS

BOOST YOUR CAREER IN DEVOPS & CLOUD

SKILLS TODAY FOR A SMARTER TOMORROW

JENKINS, ANSIBLE & MAVEN
DEVOPS AUTOMATION TOOLS

JENKINS — CI/CD AUTOMATION

JENKINS FUNDAMENTALS
- Jenkins Installation
- Jenkins Configuration
- Jenkins Architecture
- Controllers & Agents
- Executors
- Jobs & Build History

JENKINS PIPELINES
- Freestyle Projects
- Pipeline as Code
- Jenkinsfile
- GitHub & GitLab Integration
- Automated Build Triggers

BUILD AUTOMATION
- SCM Polling
- Webhooks
- Scheduled Builds
- Maven Integration
- Automated Testing
- Build & Deployment

SECURITY & ADVANCED PIPELINES
- User Management
- Roles & Permissions
- Credentials Management
- Secret Management
- Parallel Stages
- Conditional Builds
- Pipeline Integration

Jenkins + Git + Docker + Ansible + Slack

ANSIBLE — AUTOMATION

ANSIBLE FUNDAMENTALS
- Ansible Installation
- Control Node
- Managed Nodes
- Inventory Files
- Playbooks
- Roles & Tasks

CONFIGURATION MANAGEMENT
- Ansible Playbooks
- Tasks & Handlers
- File Module
- User Module
- apt & yum Modules
- Application Automation

SECURITY & ADVANCED ANSIBLE
- Ansible Vault
- Secure Variables
- Static Inventory
- Dynamic Inventory
- Ansible Galaxy
- Reusable Roles
- Multi-Node Orchestration

MAVEN — BUILD AUTOMATION

MAVEN FUNDAMENTALS
- Maven Architecture
- Project Object Model — POM
- Maven Lifecycle
- validate
- compile
- test
- package
- install
- deploy

PROJECT BUILDING
- Dependency Management
- pom.xml Configuration
- Java Project Builds
- JAR & WAR Packaging
- Maven Plugins

ADVANCED MAVEN
- Compiler Plugin
- Surefire Plugin
- Deploy Plugin
- Multi-Module Projects
- Maven Profiles
- Environment-Based Builds
- Jenkins Integration



BUILD MODERN DEVOPS PIPELINES

AUTOMATE WITH CONFIDENCE

SECURE YOUR INFRASTRUCTURE

DEPLOY FASTER, ACHIEVE MORE

SKILLS TODAY FOR A SAFER TOMORROW

KEY SKILLS

Plan → Configure → Build → Deploy → Monitor → Optimize

DOCKER, PROJECTS & CAREER
CONTAINERIZATION & CLOUD DEVOPS

10. DOCKER — CONTAINERIZATION

DOCKER FUNDAMENTALS
- Containers vs Virtual Machines
- Docker Architecture
- Docker Installation
- Docker CLI
- Docker Images
- Docker Containers

DOCKER COMMANDS
- docker run
- docker ps
- docker images
- docker exec
- docker start
- docker stop
- docker restart

DOCKER IMAGES & CONTAINERS
- Dockerfile
- Image Building
- Container Lifecycle
- Image Management
- Application Deployment

DOCKER NETWORKING & STORAGE
- Bridge Network
- Host Network
- Overlay Network
- Docker Volumes
- Persistent Storage
- Multi-Container Applications

DOCKER COMPOSE
- Docker Compose Fundamentals
- docker-compose.yml
- Multi-Service Applications
- Service Configuration
- Application Orchestration

CONTAINER ORCHESTRATION

Docker Swarm
- Cluster Setup
- Services
- Scaling

Kubernetes — Introduction
- Container Orchestration
- Kubernetes Architecture
- Pods & Services
- Basic Deployment Concepts

HANDS-ON PROJECTS
BUILD REAL-WORLD CLOUD & DEVOPS PROJECTS

AWS Cloud Deployment
Deploy and manage a scalable application using EC2, S3, VPC, IAM and Load Balancing.

CI/CD Pipeline
Build an automated pipeline using Git, Jenkins, Maven and Docker.

Infrastructure Automation
Automate server configuration and application deployment using Ansible.

Dockerized Application
Containerize a real-world application and manage it using Docker Compose.

Cloud Monitoring Solution
Implement application monitoring and logging using AWS CloudWatch and CloudTrail.

SKILLS YOU WILL MASTER
- Linux Administration
- AWS Cloud
- Cloud Networking
- Cloud Security
- Git & GitHub
- DevOps
- CI/CD
- Jenkins
- Ansible
- Maven
- Docker
- Docker Compose
- Kubernetee Fundamentals
- Cloud Automation

CAREER OPPORTUNITIES
- Cloud Engineer
- AWS Cloud Engineer
- DevOps Engineer
- Cloud DevOps Engineer
- Linux System Administrator
- AWS Administrator
- Site Reliability Engineer
- CI/CD Engineer
- Automation Engineer
- Cloud Support Engineer
- Infrastructure Engineer
- DevOps Automation Engineer

BUILD MODERN SKILLS

DEVELOP SCALABLE SOLUTIONS

GAIN PRACTICAL SKILLS

BECOME INDUSTRY-READY

LEARN TODAY
SECURE TOMORROW